

A Continuous Manufacturing Line Generating Organozinc Species in Flow: Enhancing the Simmons-Smith Reaction Including Post-Reaction Processing

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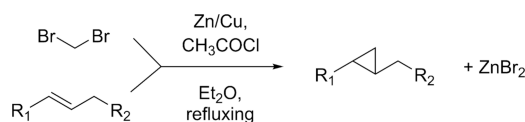
ABSTRACT: The cyclopropyl fragment is paramount in the pharmaceutical industry, as a wide range of active pharmaceutical ingredients inhere this versatile molecular structure. Recently, even higher importance has been attributed to the cyclopropane ring since many transition drug candidates utilize this carbon–carbon arrangement's beneficial features. A reliable way to produce cyclopropanes is via the Simmons-Smith reaction, which utilizes a zinc–copper couple to transfer the methylene group of a dihalomethane to the double bond of a target molecule. The whole process involves a quenching and filtration step and is usually completed by a time-consuming extraction of metal salts and other impurities. This work presents a reliable and reproducible method to perform all unit operations continuously. It includes the synthesis step in a packed-bed reactor filled with a zinc–copper couple, activated in flow, eliminating the need for a filtration unit. It is followed by a two-stage cross-flow extraction for simultaneous quenching and removal of zinc salts from the product solution. Finally, the extraction efficiency is monitored atline by an in-house-developed method using chromophoric reagents in continuous flow.

KEYWORDS: continuous, flow chemistry, Simmons-Smith, cyclopropanation, organozinc, Zn/Cu, zinc–copper couple, membrane, extraction, chromophore, PAT, process analytical technology, process intensification, Bayesian optimization, at-line analysis, UV–vis, real-time

INTRODUCTION

Scientists worldwide are screening for new potential drug candidates, comprising beneficial molecular scaffolds that facilitate activating or inhibiting specific enzymes or receptors without unwanted side effects.^{1,2} A recurrent molecular structure is the cyclopropyl ring. It increases the sp³-hybridized character and nonplanar attributes in potential drug entities.³ As a molecular substituent, its short carbon–carbon bonds⁴ and the strength of its carbon–hydrogen bonds⁵ make the cyclopropyl ring appealing regarding increased potency, metabolic stability, and reduced off-target effects.⁶ The introduction of a cyclopropyl ring into a molecular scaffold can be done in various ways, whereby the Simmons-Smith reaction is considered the most reliable and often-used method.^{7–9} Simmons and Smith discovered in 1958, that the organozinc intermediate, formed by mixing diiodomethane with copper-activated zinc, Zn(Cu),¹⁰ in refluxing ether, could be utilized to convert various olefins into their corresponding cyclopropanes.¹¹ Since then, several alternatives have been reported in literature: In the Furukawa modification reported in 1968,¹² the zinc–copper couple is replaced with dialkylzinc, and in the variant of Charette et al.¹³ metal alkoxides are used as efficient chiral controllers to yield cyclopropyl derivatives of allylic alcohols in high enantiomeric excess. Friedrich et al. reported a synthetic variant (Scheme 1) of the classical Simmons-Smith reaction, using zinc dust, copper chloride, and acetyl chloride to produce the couple in situ.¹⁴

Scheme 1. Simmons-Smith Cyclopropanation Regarding Friedrich et al



Although further variants and modifications of the Simmons-Smith reaction exist, only Nova-Fernández et al. explored the potential of transferring the reaction into a continuous process, by utilizing an in-batch-produced zinc–copper couple filled in a perfluoroalkoxy alkane (PFA)-tube reactor.¹⁵ The general advantages and disadvantages of continuous processes in pharmaceutical applications have been discussed thoroughly before.^{16–20} Here, we want to highlight the benefits associated with a continuous Simmons-Smith synthesis by comparing it to the classical variant, performed in batch, using zinc dust and a copper salt. One of the challenges in handling Simmons-Smith reactions in batch has been the highly exothermic nature of the

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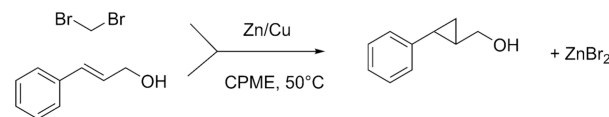


reaction, accompanied by strong ethylene formation. In particular, the scale-up of the reaction to an industrial scale is a hazardous operation. Therefore, heat transfer limitations are counteracted by slowly metering the dihalomethane to the reaction solution, containing the alkene, over at least several hours.^{14,21} Subsequently, highly active, pyrophoric zinc particles need to be separated from the product solution by filtration. When quenching the reaction solution with ammonium chloride, the acidity of the salt is sufficient to further remove still passivated areas of the zinc particles, which are mainly comprised of basic zinc carbonate, $\text{Zn}_5(\text{OH})_6(\text{CO}_3)_2$, and zinc oxide,²² therefore keeping it active. Hence, filtration is performed under inert conditions, the filter cake is then stored in silicon oil. After filtering the metal particles from the quenched organic phase, a considerable amount of metal ions, especially zinc bromide, remains present in the solution. Therefore, an additional purification step to lower the zinc salt concentration is needed. Liquid–liquid extraction (LLE) is a common unit operation to remove metal salts from the product solution and can be conducted in extraction columns, mixer settler combinations, or by using a series of batch extractors. The concentration of the metal salt is then commonly determined by offline analysis such as atomic absorption spectroscopy (AAS) and inductively coupled plasma spectroscopy (ICP-OES/MS).²³

Changing the operational mode from batch to a continuous flow process eliminates the need for the filtration step and allows the implementation of in- and online measurements. Moreover, monitoring critical process parameters (CPPs) allows immediate response if a product is outside the desired limits, e.g. by adjusting the flow of the wash solution in the extraction of metal salts. Therefore, we present a new and fully continuous approach to the Simmons–Smith reaction including reaction and workup, overcoming limitations, and potentially hazardous operation steps associated with procedures described in prior art. Our improvements involve the activation of zinc dust and the synthesis of the reactive metal couple in flow. We use empty HPLC columns filled with zinc dust and copper salt as packed bed reactors. They are robust, pressure-resistant, show excellent heat transfer properties, and can accommodate powders with particle sizes ranging from mm down to μm . These attributes make them ideal tools for performing heterogeneous small-scale reactions in flow.^{24–26} Cyclopentyl methyl ether (CPME), a green solvent,²⁷ was selected for the reaction as it typically provides higher yields in the Simmons–Smith reaction compared to other ethereal solvents.²⁸ Furthermore, as an alternative to the economically unfavorable diiodomethane, dibromomethane (DBM) was used. To optimize the reaction yield, instead of a one-variable-at-a-time approach (OVAT), which suffers from significant drawbacks in the optimization of chemical reactions,^{29,30} the implementation of a Bayesian optimization (BO) algorithm via the *Python* library *skopt* was conducted. Implementing optimization algorithms for chemical reactions is versatile and well-documented.^{31–34} BO, well suited for noisy or expensive black-box models^{35–37} (e.g., high evaluation cost of data points), relies on a probabilistic model, typically a Gaussian process, to create a posterior distribution over the unknown objective function based on collected data points. In each iteration of BO, an acquisition function is used to determine the best next point in the design space. In each iteration, when new data (i.e., the reaction yield) is available, the GP is updated, resulting in an improved model of the

acquisition function and, therefore, an improved optimization process via a suggestion of new values for a parameter set.³⁸ Generally, this approach greatly reduces the number of experimental trials required, therefore being considerably more efficient compared to process optimization by a full factorial design of experiments (DoE). As a model substrate for the Simmons–Smith cyclopropanation, cinnamyl alcohol was used (Scheme 2). This alkene was chosen because it is

Scheme 2. Cyclopropanation of Cinnamyl Alcohol with Dibromomethane in CPME



nontoxic, cheap to acquire, and commercially available in high purity. Further publications that used cinnamyl alcohol to showcase different cyclopropanation variants in batches exist; very high to excellent yields were achieved.^{8,39} The only continuous approach so far used a zinc–copper couple filled in a PFA tube and was also able to achieve excellent yields.¹⁵ The present work aims to introduce considerable improvements to prior art while still achieving excellent yields by activating the zinc–copper couple in flow, which has not been shown before, using a green solvent and the economically more favorable DBM. Further, we demonstrate a downstream microfluidic extraction unit comprising two stages of 3D-printed static mixers and membrane separators, reducing the metal salt concentration below 100 mg/L. The extraction process is monitored by an analytical unit, combining flow equipment such as static mixers and membrane separators, with a colorimetric method for Zn^{2+} determination, based on the chromophore dithizone.

EXPERIMENTAL SECTION

Preparation of the Packed Bed. A commercially available empty stainless steel HPLC column (I.D. $\times$ L, 8 $\times$ 125 mm) with a free internal volume of 6.28 cm^3 was covered at the bottom with a metal frit (5 μm pore size), PTFE O-ring, ferrule, and the external column fitting. The column was filled with zinc dust (97%, particle size $< 65 \mu\text{m}$, Sigma-Aldrich). The method of LeGoff et al.⁴⁰ served as a starting point for the present work, but several modifications were implemented to realize the continuous approach. In the first attempts using only zinc dust and copper(II)acetate monohydrate ($\text{Cu}(\text{OAc})_2 \times \text{H}_2\text{O}$, 98%, Sigma-Aldrich), regular clogging of the column and high pressures were observed. Therefore, chemically inert silicon carbide (200 mesh, Sigma-Aldrich) was added to the packed bed in a volumetric ratio of 1:1 to the metal/metal salt blend, which resolved the issue. Packed bed: First, 0.3 g of silicon carbide was filled into the HPLC column to prevent the zinc/copper salt mixture from getting in contact with the metal frit. Ground $\text{Cu}(\text{OAc})_2 \times \text{H}_2\text{O}$, zinc dust, and silicon carbide were accurately weighed and thoroughly mixed using a mortar and pestle. The proportion of copper relative to the total weight of the metallic mixture was varied in increments of 2.5 wt % between 0 to 10 wt % during optimization runs. Then, the total volume of the mixture was combined with silicon carbide in a 1:1 volumetric ratio. The mixture containing 7.50 g of zinc, allowing consistent comparison across experiments, was then carefully loaded into the column. The column was

not tapped to avoid compaction of the bed, which might lead to irregular flow behavior or blocking. The remaining empty part of the column was again filled with pure silicon carbide.

Then, the top of the column was covered as described for the bottom part. Afterward, a 3D-printed mount was placed in a water bath maintained at 50 °C by an IKA C-MAG HS7 heating plate. Then, the assembled column was fully immersed in the stirred water bath and placed in the 3D-printed holder, facilitating exact vertical positioning (Figure 1). Subsequently,

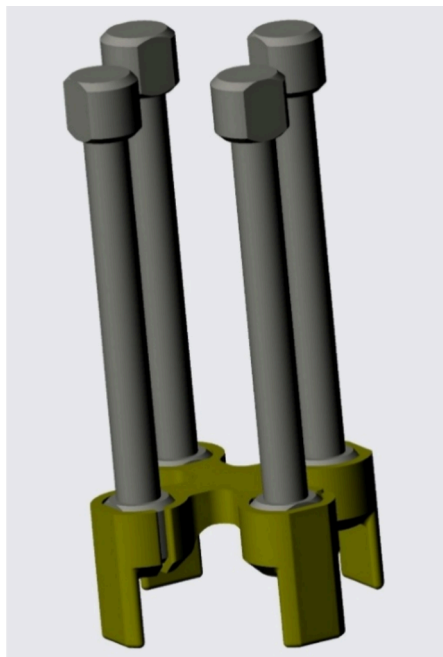


Figure 1. 3D-printed mount with HPLC columns for parallel operation of up to four columns.

the column was connected to a backpressure regulator (BPR), (Zaiput BPR-10) using appropriate standard PEEK fittings and PTFE tubing (I.D. $\times$ O.D., 1/32 in. $\times$ 1/16 in.), keeping the tubing as short as possible.

Activation of Zinc and Flushing of the Column. Activating zinc dust in a column with $\text{Cu}(\text{OAc})_2 \times \text{H}_2\text{O}$ dissolved in acetic acid has been described as a feasible approach for batch processing.⁴⁰ During initial investigations, a similar approach transferred to continuous processing resulted in an uneven Zn(Cu) formation during the progress of the solution front through the column. Therefore, $\text{Cu}(\text{OAc})_2 \times \text{H}_2\text{O}$ was added directly to the zinc powder as described above.

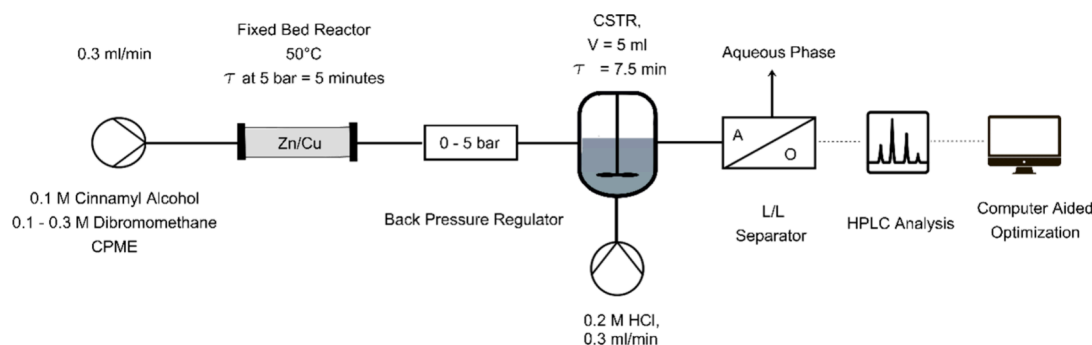
The activation solution containing 0.2 M acetyl chloride (98%, Sigma-Aldrich), 0.1 M DBM (99%, Sigma-Aldrich) in 2-methyltetrahydrofuran (2-Me-THF), (>99%, Carl Roth), was pumped at a consistent flow rate of 0.5 mL/min for 20 min through the column using a 20 mL Braun Injekt Luer Lock syringe (all polypropylene (PP)), coupled with a VIT-FIT HP, Lambda Instruments syringe pump. Concurrently, a back-pressure regulator with 1.5 bar was applied to the outlet of the column to increase the residence time of the activation solution in the column, which otherwise would have been affected by the formation of ethylene gas. The reactor bed itself did not induce significant pressure loss (<0.1 bar).

During activation, the acyl chloride combines with the hydroxylic impurities of the zinc surface as well as with the water originating from the copper source to form hydrochloric and acetic acid. Both are beneficial in the formation of the Zn/Cu couple as shown by LeGoeff and Friedrich.^{14,40} In addition, DBM is believed to create microscopic holes in the zinc particles that further activate the zinc by increasing its surface area. The successful synthesis of the Zn/Cu couple was observed by the transparent color of the solution leaving the column, indicating the absence of $\text{Cu}(\text{OAc})_2$. Furthermore, strong gas formation was observed. Following the activation step, degassed cyclopentyl methyl ether (CPME, >99%, Carl Roth) was pumped through the column at a flow rate of 0.5 mL/min for 5 min. This step served to remove residual 2-Me-THF and acetyl chloride and to prime the reactor bed for the subsequent synthesis.

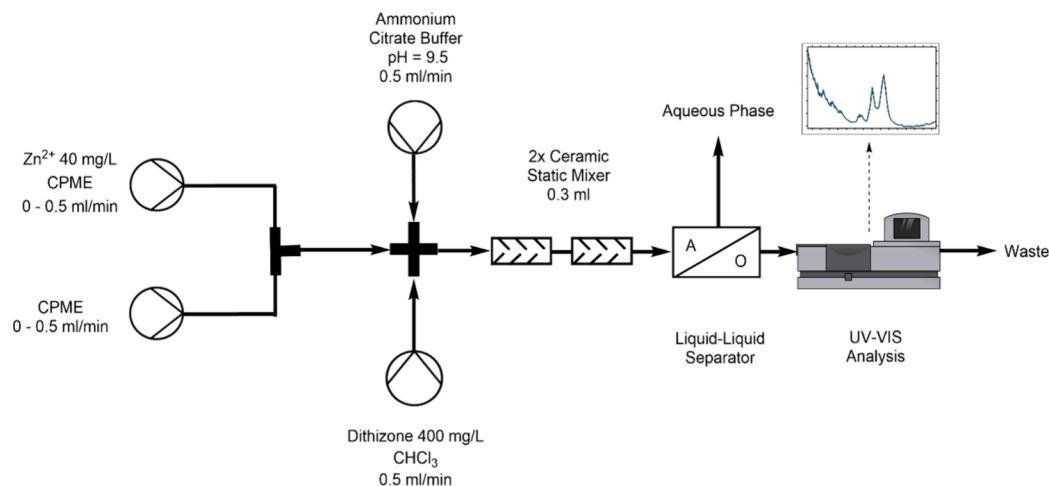
Optimization of the Simmons-Smith Reaction. Preparation of an exemplary reaction solution (0.1 M DBM, 0.1 M cinnamyl alcohol): 0.68 M (0.01 mol) of DBM, 1.29 mL (0.01 mol) of cinnamyl alcohol and 0.25 mL (2 mmol) of toluene as a tracer were pipetted in a 100 mL oven-dried volumetric flask, then filled up to 100 mL with CPME. Then the mixture was transferred into a 100 mL screw cap bottle filled with 10 g of 3 Å molecular sieves. After 48 h, the mixture was degassed by applying a vacuum (100 mbar) and ultrasound for 30 s. The remaining air in the bottle was exchanged with an argon atmosphere and was now ready to use.

Several reaction solutions were prepared for optimization purposes to explore different conditions within the experimental design space. These solutions differ solely in the molar concentration ratio of DBM (1–3 mol equiv) to cinnamyl alcohol (1 mol equiv) while maintaining constant concentrations of toluene (0.2 mol equiv), which acted as an inert tracer to account for errors during sampling and HPLC analysis.

Scheme 3. Experimental Setup Used for the Optimization of the Simmons-Smith Synthesis



Scheme 4. Process Flow Diagram of the Setup That Was Used for the Calibration of Zinc Measurements in Flow



Exemplary Optimization Run. The reaction solution was pumped through the activated packed bed with 0.3 mL/min for 60 min using a 20 mL *Braun* Injekt Luer Lock syringe (PP) on a VIT-FIT HP, *Lambda Instruments* syringe pump. After the reactor, the product solution entered the BPR. Its back pressure was regulated by a connection to an argon line, with adjustable pressure, different pressures could be set during the experiment. The applied backpressure values were validated by integrating a Honeywell ABP series pressure sensor connected by a T-piece and a fluoroelastomer capillary (FKM, I.D. $\times$ O.D., 1/32 in. $\times$ 1/16 in.) to the setup.

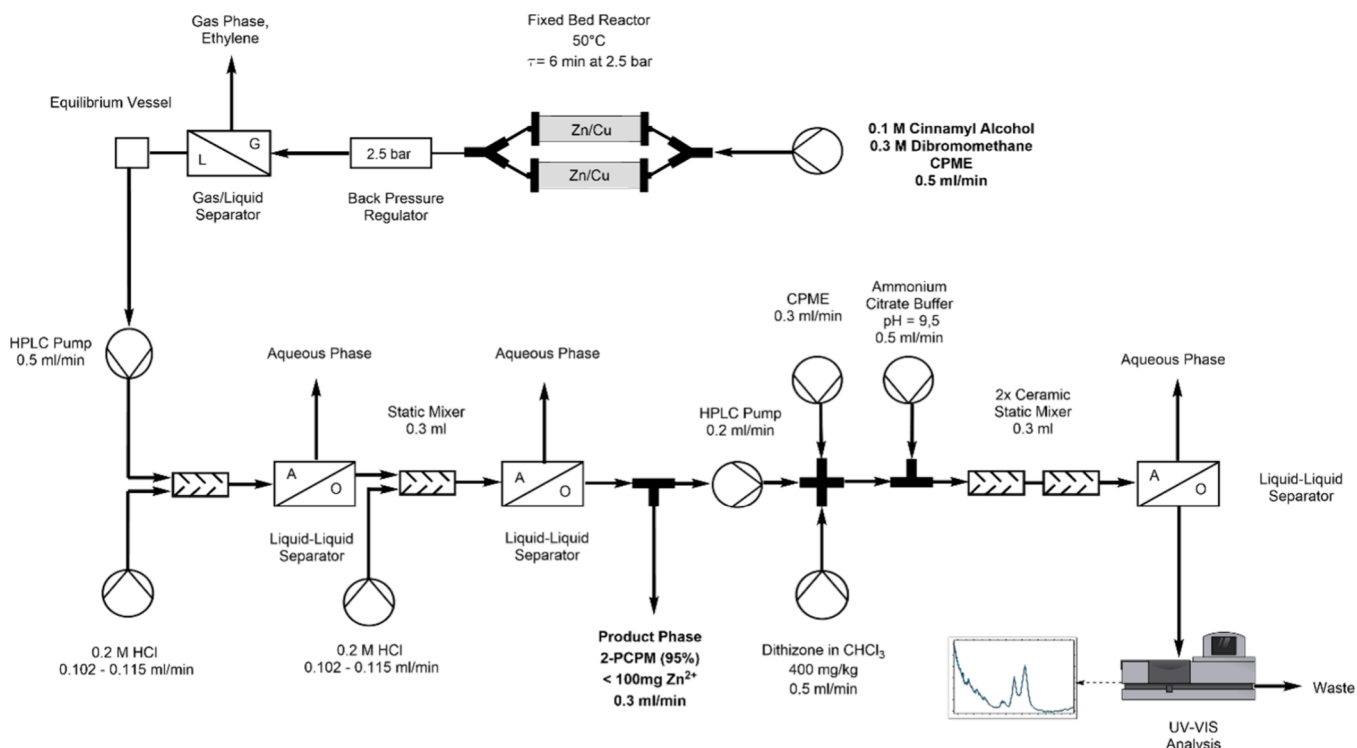
After the BPR, the product solution entered a 3D-printed continuous stirred tank reactor (CSTR) with an internal volume of 5 mL (Scheme 3). The reaction was quenched continuously inside the CSTR using 0.2 M HCl. The biphasic mixture was then continuously separated in a *Zaiput* SEP-10 membrane separator equipped with an OB-2000 membrane. All other tubing, except the connection to the pressure sensor, was realized using appropriate standard PEEK fittings and PTFE tubing (I.D. $\times$ O.D., 1/32 in. $\times$ 1/16 in.), keeping the tubing as short as possible. The organic phase, which was collected between 30 and 60 min after the start of the experiment, was then dried over sodium sulfate. The yield (2-phenylcyclopropyl)methanol (2-PCPM) was quantified via HPLC analysis (Supporting Information (SI)).

Bayesian Optimization of the Simmons-Smith Synthesis. Three distinct parameters and their effect on the reaction yield were investigated in detail: (A) The amount of copper salt $\text{Cu}(\text{OAc})_2 \times \text{H}_2\text{O}$ in relation to zinc (0–10%, w/w), (B) the molar ratio of DBM to the alkene in the reaction, and (C) the backpressure in the reactor (0–5 bar relative to atmospheric pressure). Therefore, these factors, which have not been closely investigated for a flow approach before, have been studied through five discrete values each, which were uniformly distributed in the parameter space, resulting in 125 parameter sets in total. Before the optimization started, five experiments of the starting point were conducted as described in the experimental section “Exemplary optimization run”. These experiments determined the standard deviation of the process yield at the central point of the design space (5% $\text{Cu}(\text{OAc})_2 \times \text{H}_2\text{O}$, (w/w), substrate ratio of 2, and 2.5 bar back pressure) to be 4.1%. The observed value was then considered in the code driving the optimization process. Latin hypercube sampling of five points was chosen to provide initial

information on the response surface of the design space. Compared to random sampling, this method ensures that each interval is sampled exactly once for each parameter. This covers a more uniform area of interest, ensuring that the design or parameter space is evenly explored.⁴¹ Then, based on the previous experiments, the algorithm proposed a new promising parameter set. As soon as the reaction yield for a new operating point has been determined by HPLC, this value is added to the BO algorithm and a new point is obtained. To increase the probability of finding the global optimum, after having determined a local optimum, the algorithm moves away to yet unexplored regions in the design space. The clustering of points 14–16 in the same region—with only minor differences in one parameter—and their consistently high yields suggest that the optimization had converged to a robust optimum. BO balances exploitation (refining close-to-optimum regions) and exploration of new regions. After the identification of a region with high yields, subsequent iterations proposed points in the design space lying in yet unexplored regions. For the present work, the optimization was stopped after 16 iterations since these exploratory moves proposed points with e.g. no copper salt, resulting in experiments with no yield, confirming that the high-yield region was a reliable optimum within the studied range.

Extraction of Zn^{2+} . The extraction of zinc salts was implemented directly after the L/L separator in the synthesis step (Scheme 3) by contacting the organic product phase with 0.2 molar HCl in a 2-stage setup. Since all the copper salt was completely consumed during the activation process, Cu^{2+} ions were not expected to be found in significant amounts in the product solution. The extraction setup consisted of two 316L stainless steel sintered, 3D-printed, static mixers⁴² and two membrane separators from *Zaiput*, type SEP-10, equipped with OB-2000 membranes. The wash phase was pumped by a dual channel syringe pump (NE-4000, *New Era pump systems*), the organic phase by an HPLC pump (*Knauer Azura P4.1S*). The mixing and the separation units were connected by standard PEEK fittings and PTFE capillaries (I.D. $\times$ O.D., 1/32 in. $\times$ 1/16 in.) keeping the tubing as short as possible. Compared to ethereal solvents like tetrahydrofuran (THF) or 2-Me-THF, CPME is less polar. Consequently, zinc bromide preferentially accumulates in the aqueous phase, as indicated by its partition coefficient, thereby reducing the amount of wash medium required.

Scheme 5. Setup for the Continuous Simmons-Smith Process with Work-Up Comprising the Synthesis and Extraction Step as well as an Analytical Unit



Zinc Analysis in Flow. The method developed to quantify the concentration of Zn^{2+} ions in continuous flow was based on the provided protocol of the United States Pharmacopeias (USP) 2024 zinc determination in batch, using a dithizone solution in chloroform (10 mg/L) as a chromophore.⁴³ As reported in the USP, using a 10 mg/L dithizone solution and a cuvette with a 10 mm path length, Zn^{2+} up to 1.0 mg/L can be quantified. To widen the measuring range, the concentration of the chromophore solution was increased to 40 mg/L, and a Hellma quartz glass flow cell with 0.5 mm path length was used, ensuring the sensitivity of the method in a Zn^{2+} concentration range between 2 to 20 mg/L (1:2 dilution of chromophore-carrying phase with calibration solution). To calibrate the method, three syringe pumps were used. The first syringe (VIT-FIT HP, *Lambda Instruments*) was loaded with a solution of zinc bromide in CPME containing 40 mg Zn^{2+} ions per liter. The second syringe (VIT-FIT HP, *Lambda Instruments*) was loaded with pure CPME. The concentration of the solution was adjusted by mixing the two solutions in 20% increments regarding the maximum concentration, using a T-piece, keeping the total flow rate always at 0.5 mL/min. The calibration solution is then combined in a cross-channel mixer with the chromophore-carrying phase (1:2, which reduces the Zn^{2+} concentration of both combined phases by 50%) as well as an 0.11 M aqueous ammonium citrate buffer (buffer stock solution diluted (SI), diluted 1:20, pH 9.5), pumped by a dual syringe pump (NE-4000, *New Era pump systems*). For all solutions, a 20 mL *Braun Injekt Luer Lock* syringe (PP) was used. Mixing was accomplished using two consecutively installed, 3D-printed static mixers (printed by *Lithoz, Vienna, Austria*) sintered from ceramic powder (**Scheme 4**). The geometry of the mixers is identical to the geometry of the mixers used in the extraction process. Before entering the flow cell, the organic phase was separated from the aqueous phase

using a *Zaiput* SEP-10 liquid–liquid separator; absorbance was determined at 525–527 nm using a UV–vis spectrometer (*Avantes* AvaLight-DS-DUV with deuterium lamp detector: *Avantes* AvaSpec-ULS2048). All units were connected by standard PEEK fittings and PTFE capillaries (I.D. $\times$ O.D., 1/32 in. $\times$ 1/16 in.), keeping the tubing as short as possible. All contacted surfaces had to be inert regarding the chromophore.

Setup for Continuous Processing. The complete setup of the continuous production line including synthesis, extraction, and analysis is depicted in [Scheme 5](#). All pumps, connections, and other necessary materials used for this setup are already described in chapters 2.1–2.4 and kept the same if not stated otherwise. Since a considerable fraction of the product solution was required for Zn^{2+} concentration measurements, the total throughput of the setup was increased by using two packed-bed reaction columns in parallel. Increasing the column diameter or length would be an alternative option for scale-up. Small deviations in the flow velocity through each column are possible but considering a back pressure of 2.5 bar and assuming only small pressure losses in the column (<0.1 bar), those deviations can be considered negligible. An equilibrium vessel (HPLC vial, $V = 2$ mL) was installed to connect the synthesis step of the setup to the downstream workup section, buffering the discontinuous suction of the downstream HPLC pump (*Knauer* AZURA P 4.1S). Then, the product solution entered the two-stage cross-current extraction unit, and subsequently, the T-piece, where a side stream of the extracted phase was withdrawn. This side stream is used for the analytical unit and was pumped by a *Knauer* AZURA P 4.1S HPLC pump. The internal volume of the pump was determined to be only 0.3 mL, which minimizes the dead volume of the analytical unit. By diluting the organic phase with pure CPME 2:5 (v/v), the analytical range was extended to reliably measure a concentration range from 10 to 100 mg/

L. The concentration of zinc was determined from the absorbance measurements by UV–vis spectroscopy as described in the section “zinc analysis in flow”.

RESULTS AND DISCUSSION

Finding the Optimal Process Conditions with BO. An overview of the yield of 2-PCPM is given in Figure 2. Based on

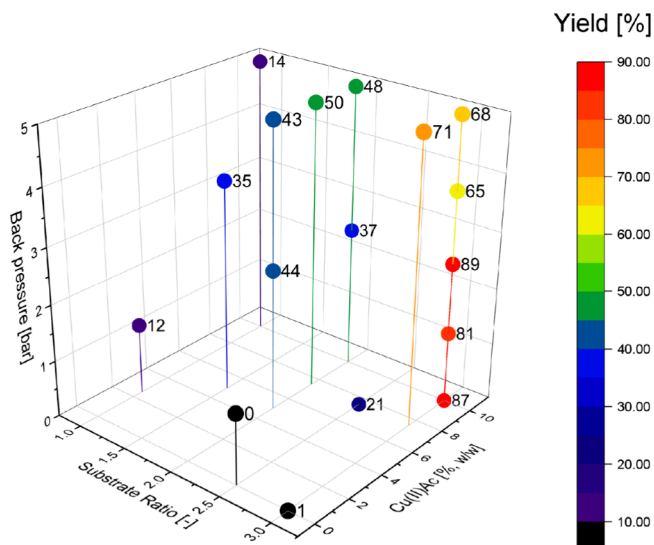


Figure 2. Yield of 2-PCPM as a function of the weight percentage of copper acetate in the column, substrate ratio and back pressure on the column.

the analysis of the experimental data, two parameters were identified that are heavily influencing the reaction outcome. At iterations 4 and 13, no copper salt was added to fill the column (A detailed table of process parameters, the conversion of

DBM and cinnamyl alcohol, and the yield for each experiment can be found in the SI). Although the zinc dust is activated (passivating layer is etched by acetyl chloride) and provides a high surface area, the data indicate that at set reaction conditions, only minor amounts (1–3%) of DBM were found to be converted; nevertheless, no product formation could be observed. The second parameter highly affecting the yield is the concentration of DBM. It is known that DBM, in comparison to diiodomethane, needs a higher molar ratio with respect to alkene,¹⁴ but this has not been investigated in flow applications before. The findings of this work suggest that the hypothesis made for batch processes can be translated to flow processes as well. In all experiments, where the molecular ratio of DBM to allyl alcohol was 3:1, $\alpha > 90\%$ conversion of DBM always led to a very good yield. In an experimental iteration with a 2-fold ratio and high copper salt concentration, only 58% conversion of cinnamyl alcohol was achieved, although nearly 98% of the dihalomethane was converted (Data for the conversion of DBM can be found in the SI). Nova-Fernández et al. reported a significant influence of back pressure on the system, where an elevated pressure of 75 psi (~ 5.2 bar) resulted in higher yields compared to operation at atmospheric pressure.¹⁵ In our flow setup using DBM as the carbon source, this finding could not be confirmed. First, the relative differences between experiments with the highest DBM ratio and copper salt content are not as pronounced as in Nova-Fernández's attempt with diiodomethane; second, here, lower backpressures resulted in higher yields. The observed data seem counterintuitive since the gas formation (mostly ethylene), which occurs as a highly pronounced side reaction in this operation, significantly reduces the mean residence time (5 min at 5 bar) of the reaction solution inside the column. The optimal operating points were found at the highest copper salt and DBM concentration, and pressures up to 2.5 bar.

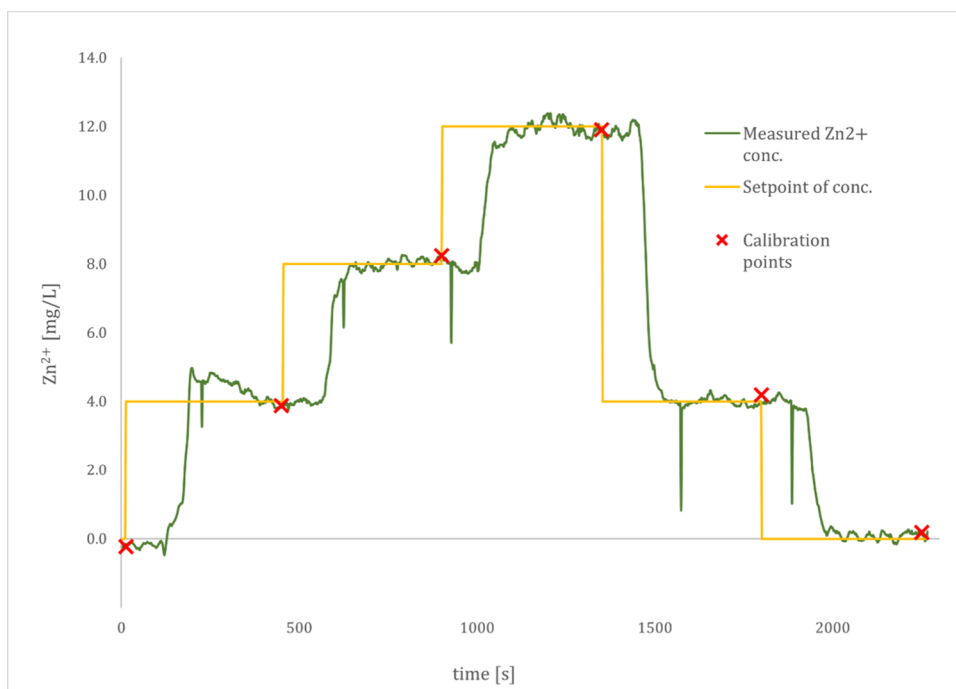


Figure 3. Results of the calibration of flow setup with zinc bromide. The last data entry at each set point is used for the calibration curve. The 2-fold dilution of the calibration solution caused by the chromophore-carrying phase is already accounted for in this graph.

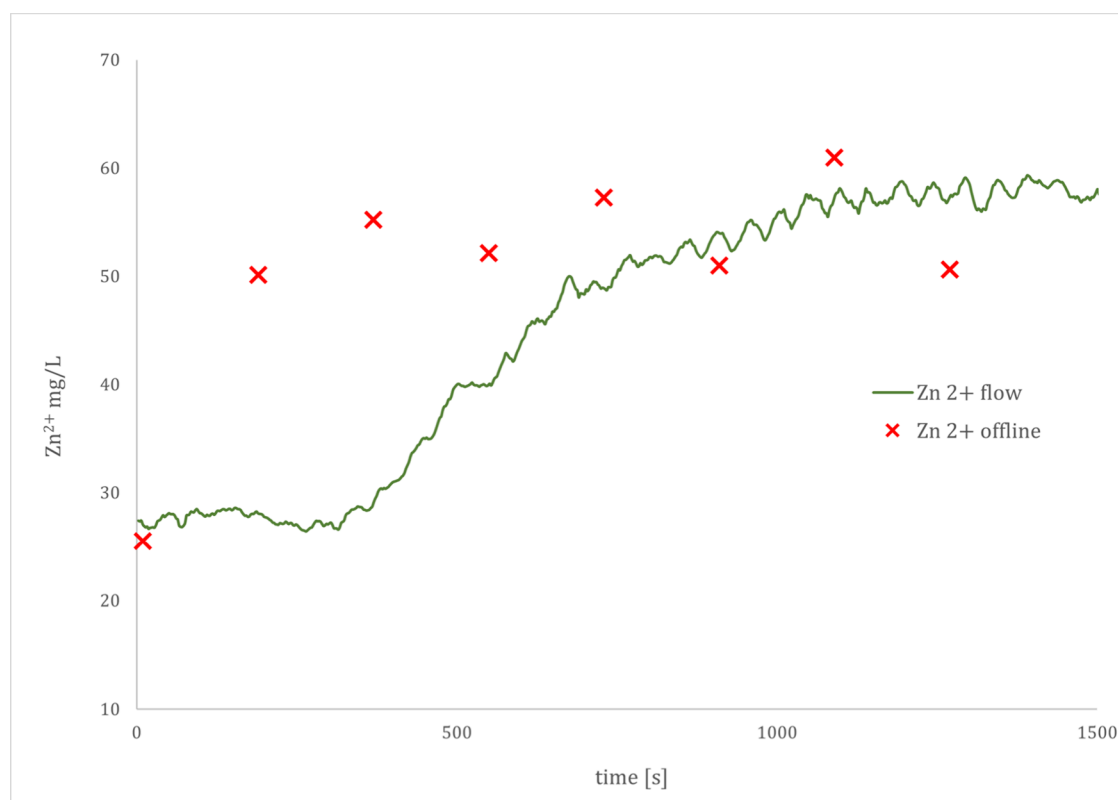


Figure 4. Offline vs atline measurement of Zn^{2+} with a dithizone-based measurement technique. At $t = 0$ the extraction ratio of wash to organic phase is increased from 1:0.46 to 1:0.41.

Scale Up of Synthesis. During the optimization procedure, it was observed that even at the best parameter sets, the conversion of DBM was not fully complete. Therefore, the total flow rate over both columns was reduced to 0.5 mL/min, resulting in a flow rate of 0.25 mL/min in each column, increasing the residence time by 20%. At this flow rate, full conversion of DBM could be achieved, increasing the yield by an additional 6–95%. The pressure loss inside the column was measured to be below 100 mbar and did not change significantly throughout the experiment (60 min), therefore, a uniform flow through both columns was assumed. Further scale-up would require additional parallelization or the use of longer columns. Increasing the volume of the reactor bed by using a wider column requires further evaluation of heat transfer limitations. Additionally, although not observed in our experimental runs, a possible blockage of the frits caused by decreasing zinc particle size must be considered. An example calculation for the continuous synthesis of a precursor of Ropanicant (a substrate for the Simmons-Smith reaction) shows that already ten HPLC columns in the size of 10×250 mm would be able to produce the intermediate at 1.37 mol/d (equal to 152 g/d Ropanicant or about 3400 doses, assuming a total yield of the final API through further synthesis steps of 50%).^{44,45}

Zn^{2+} Extraction and Quantification. A full conversion of DBM led to a concentration of Zn^{2+} of 23 g/L. Batch extraction trials revealed that a ratio of 0.4 parts of aqueous wash solution (0.2 M HCl) to 1 part of organic phase was needed to meet the target concentration of $\text{Zn}^{2+} < 100$ mg/L; via HPLC analysis, the product loss to the washing phase was determined to be $< 0.5\%$. With possible further synthesis steps

that require, e.g., solvent changes^{45,46} in mind, lowering the concentration to this set specification seemed sufficient.

As a first performance test of the proposed analytical unit, a calibration of the experimental setup as described in 2.4 was performed. A linear and fast response to the concentration changes (Figure 3) has been observed. The data points used for the calibration of the flow setup were gathered when the set point of the concentration was changed. The absorbance readings at 0 mg/L Zn^{2+} and 4 mg/L Zn^{2+} were averaged. A calibration curve for Zn^{2+} with [A.U. per mg], 0.597 at 525–527 nm with $R^2 > 0.99$ was obtained. After the calibration, the analytical unit was added to the two-stage extraction setup. To demonstrate the sensitivity and capability of the proposed continuous colorimetric method, a realistic scenario with slightly differing extraction ratios was conducted. At $t = 0$, a change in the ratio of the flow rate of the washing solution to the organic phase from 0.46:1 to 0.41:1 was conducted (Figure 4). The data gathered shows that a change in the concentration of Zn^{2+} was indicated after 350 s, where the measured concentration in the product stream sharply increased as expected for a smaller extraction ratio. Offline analyses (SI) of samples that were drawn every 180 s after the T-piece following the second extraction unit were in good agreement with the atline measurement. However, a much faster increase in the concentration of Zn^{2+} was indicated. This divergence can be explained by the additional dead volume the organic phase has to pass before going through the flow cell (compared to the sampling point of the offline measurement). This assumption was backed by tracer experiments that were conducted to investigate the mean residence time distribution (RTD) of the sole extraction unit and the extraction unit combined with the analytical unit. A 3.5 min difference

between the mean residence time of the setup without and the setup with the analytical unit was revealed. The full data set of the RTD measurements and a description of how those tracer experiments were conducted are given in full detail in the SI.

CONCLUSIONS

The Simmons–Smith synthesis is a valuable method for converting C=C double bonds into cyclopropanes. In this work, we present a robust approach to transition the reaction, including workup, from multiple batch processes to a continuous line. We demonstrated that a packed-bed reactor filled with a zinc–copper salt blend, activated in flow to generate the reactive metal couple, efficiently produces reactive organozinc intermediates using the economically favorable DBM as the starting material. The continuous cyclopropanation reaction was optimized through Bayesian optimization, utilizing algorithm-driven design space exploration by varying three yield-influencing parameters (copper salt content (w/w), backpressure, and substrate ratio) to achieve excellent yields. Following the synthesis, a two-stage, cross-current extraction setup with membrane separators effectively quenched the reaction and reduced Zn²⁺ concentration by 99.5% (from 23 g/L to <100 mg/L) using only 0.4 parts of aqueous wash phase per part of organic product solution. The extraction process was validated through an in-house developed method with dithizone as a chromophoric reagent. In summary, our findings suggest that the newly introduced process scheme, considered in its entirety, is a significant alternative to the time-consuming series of batch processes, including synthesis, quenching, filtration, extraction, and analytical validation of the extraction process. Therefore, the approach presented is a step toward facilitating the transition to more efficient, continuous processes targeted by the industry.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.oprd.5c00230>.

RTD measurements, offline measurement of zinc, HPLC analysis, NMR analysis, and CAD-Files (HPLC-column holder (.stl), CSTR box (.stl), CSTR lid (.stl).) available upon request from the author (PDF)

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Author Contributions

The manuscript was written through the contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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ABBREVIATIONS

API	active pharmaceutical ingredient
LLE	liquid–liquid extraction
PFA	perfluoro alkoxy alkane
CPME	cyclopentyl methyl ether
DBM	dibromomethane
2-Me-THF	2-methyltetrahydrofuran
OVAT	one-variable-at-a-time
DoE	design of experiments
BPR	backpressure regulator
CSTR	continuous stirred-tank reactor
RTD	residence time distribution
BO	Bayesian optimization
SI	Supporting Information
HPLC	high-performance liquid chromatography
AAS	atomic absorption spectroscopy
ICP-OES/MS	inductively coupled plasma optical emission/mass spectrometry
PEEK	polyether ether ketone
PTFE	polytetrafluoroethylene
USP	United States Pharmacopeia

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